Public debt and macro trades

How do fiscal promises, monetary policy and market prices interact?

Today's route

Derive the debt ratio before debating debt sustainability.

Distinguish monetary control from fiscal and rollover constraints.

Use dated US/Japan evidence to build conditional trading scenarios.

LECTURE 7

Debt arithmetic

What makes the debt ratio rise or fall?

The government's one-period financing constraint

$$B_t = (1 + R_t^{\text{debt}})B_{t-1} - PB_t.$$

Symbol	Meaning
B_t	Nominal debt outstanding after this period
R_t^{debt}	Effective nominal interest rate on existing debt
PB_t	Primary surplus: revenues less non-interest spending

Dividing by nominal GDP

$$d_t \equiv \frac{B_t}{GDP_t^{\text{nominal}}}, \qquad pb_t \equiv \frac{PB_t}{GDP_t^{\text{nominal}}}.$$

$$\boxed{d_t = \frac{1 + R_t^{\text{debt}}}{1 + g_t^{\text{nominal}}} d_{t-1} - pb_t.}$$

g_t^{nominal}: nominal GDP growth over the same period.

The interest–growth snowball

$$\Delta d_t = \frac{R_t^{\text{debt}} - g_t^{\text{nominal}}}{1 + g_t^{\text{nominal}}} d_{t-1} - pb_t.$$

Interest above growth pushes the ratio up.

A primary surplus pushes the ratio down.

A complete debt-ratio calculation

$$d_{t-1} = 100\%, \quad R^{\text{debt}} = 6\%, \quad g^{\text{nominal}} = 3\%, \quad pb = 1\%.$$

$$\Delta d = \frac{0.06 - 0.03}{1.03} \times 1 - 0.01 = 1.91\% \text{ of GDP.}$$

The debt ratio rises from 100% to about 101.91%.

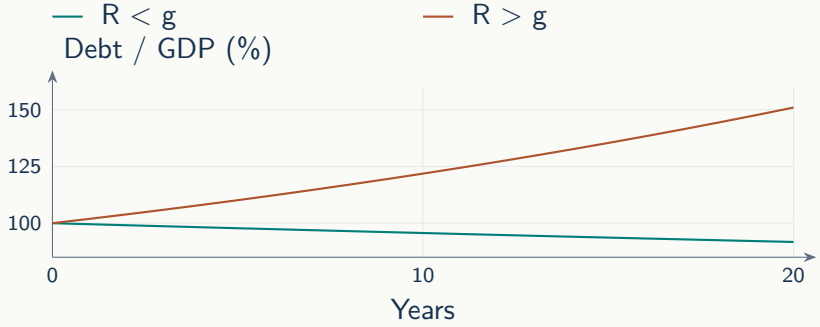
Growth above interest can permit a primary deficit

$$R^{\text{debt}} = 4\%, \quad g^{\text{nominal}} = 5\%, \quad d_{t-1} = 100\%.$$

$$pb^{\text{stabilising}} = \frac{0.04 - 0.05}{1.05} \times 1 = -0.95\% \text{ of GDP.}$$

A conditional arithmetic result. Financing risk and future rates remain.

Small parameter differences compound



Invented constant paths: green pb = -0.5%, red pb = +1% of GDP.

A rising ratio is a warning, not an instant default test

Fiscal adjustment can take time.

Risk depends on the distribution of future funding and primary balances.

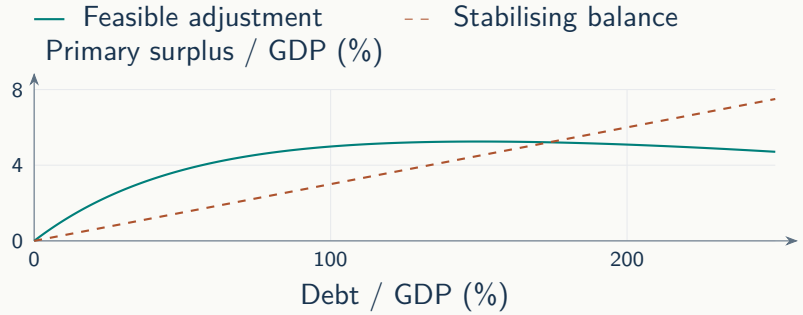
Debt definitions and the holder/currency/maturity structure matter.

A long maturity delays the cash-flow effect

Illustrative refinancing need	Share resetting this year	Rate rise
Mostly short debt	80%	+1 point
Mostly long fixed-rate debt	10%	+1 point

The same market-rate move can have very different near-term budget effects.

Fiscal fatigue and a possible adjustment limit



Invented fiscal-capacity curve and approximate interest–growth gap of 3 points.

LECTURE 7

Fiscal and monetary interaction

Can issuing money remove the government's real resource constraint?

Nominal repayment and real value are different promises

A government with its own currency can create nominal settlement balances.

It cannot create unlimited real goods by changing nominal units.

Inflation, legal constraints and political choices still affect outcomes.

Fiscal dominance

Debt-financing needs constrain the monetary response to inflation.

A central bank may face pressure to keep financing costs low.

That pressure alone does not prove which policy regime actually prevails.

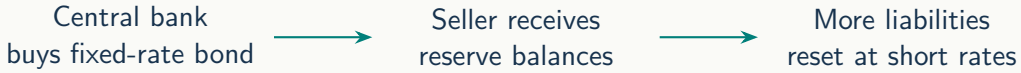
Inflation can reduce some real debt burdens

Unexpected inflation erodes the real value of fixed nominal claims.

Indexed debt and new borrowing costs can offset the benefit.

Tax/spending indexation and credibility change the fiscal result.

QE changes the public sector's liability mix



Consolidated exposure depends on reserve remuneration and the full balance sheet.

Fiscal feedback can stabilise debt in a toy model

$$pb_t = a + b\, d_{t-1}.$$

a: baseline primary balance. b: response to inherited debt.

$$d_t = \left[\frac{1 + R^{\text{debt}}}{1 + g^{\text{nominal}}} - b\right] d_{t-1} - a.$$

LECTURE 7

Dated US and Japan cases

What is verified evidence, and what is a trading hypothesis?

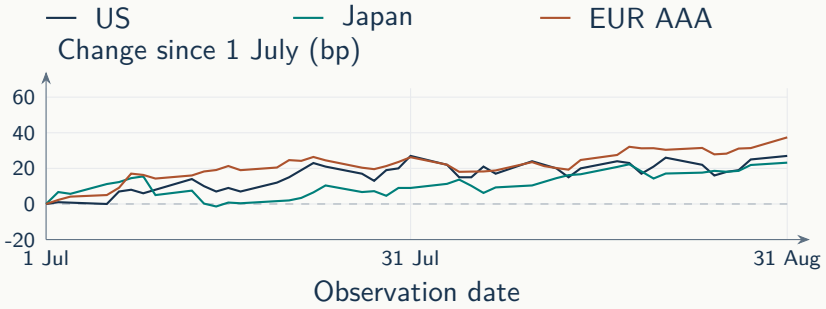
July–August 2026: a common date window

Compare 1 July with 31 August, then inspect the path between them.

Use US Treasury, Japan MOF and ECB daily series.

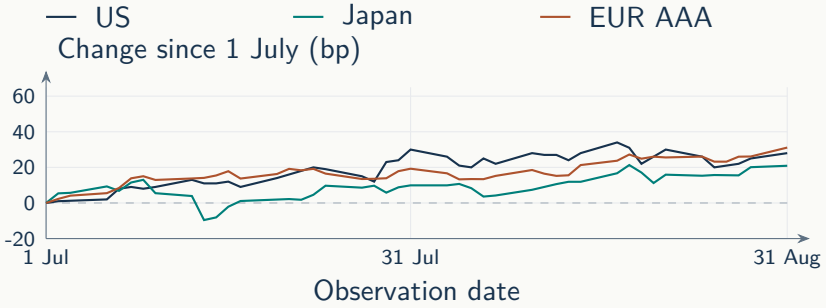
Keep currencies, yield definitions and local observation times explicit.

Summer 2026: ten-year yield changes



Official daily observations, own baseline change. Local closes/fixings differ.

Summer 2026: thirty-year yield changes

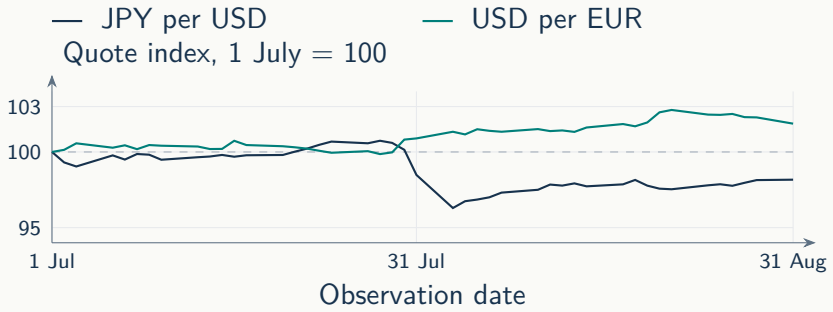


Official daily observations, own baseline change. Local closes/fixings differ.

Summer endpoints: yields rose in all three samples

Series	1 July	31 August	Change
US 10-year	4.480%	4.750%	+27.0 bp
Japan 10-year	2.711%	2.943%	+23.2 bp
EUR AAA 10-year	2.929%	3.303%	+37.4 bp
US 30-year	4.970%	5.250%	+28.0 bp
Japan 30-year	3.883%	4.092%	+20.9 bp
EUR AAA 30-year	3.379%	3.690%	+31.1 bp

Summer FX moved on its own path



Different quote directions: lower JPY per USD strengthens yen; higher USD per EUR strengthens euro.

The cross rate checks the currency story

$$\text{JPY per EUR} = (\text{JPY per USD})(\text{USD per EUR}).$$

ECB reference observation	1 July	31 August
JPY per USD, derived	162.71	159.73
USD per EUR	1.1383	1.1596
JPY per EUR	185.21	185.22

Endpoint comparison. Small net cross-rate change can hide moves within the window.

A rise in yield is not a flow-of-funds explanation

Expected real rates, expected inflation and risk premia can move together.

Hedging costs and relative duration exposure affect investor decisions.

Prices alone do not show which investors repatriated capital.

Exercise: three competing summer explanations

Candidate mechanism	What else would you check?
Higher expected future policy rates	Short-rate contracts and policy surprises
Higher duration / fiscal risk premium	Premium models, supply and risk measures
Cross-border portfolio adjustment	Hedged returns and actual transaction data

Several mechanisms may contribute. The daily charts do not choose one automatically.

US policy evidence: September 2026

On 16 September the Fed raised its target range by 0.25 points to 3.75–4.00%.

The statement described elevated inflation.

The policy decision is one input to pricing the future rate path.

US curve context: 15 October 2026

Observation	Value
Two-year Treasury par yield	4.77%
Ten-year Treasury par yield	5.10%
Ten minus two-year spread	33 basis points

Dated observations. This is a level comparison, not an identified causal decomposition.

Three hypotheses behind a higher US long yield

Hypothesis	Evidence to seek
Higher expected policy path	Dated short-rate pricing and announcements
Higher term/inflation risk premium	Models and inflation-risk indicators
More duration or funding pressure	Issuance, positioning and market functioning

Competing explanations, not an attribution of the observed snapshot.

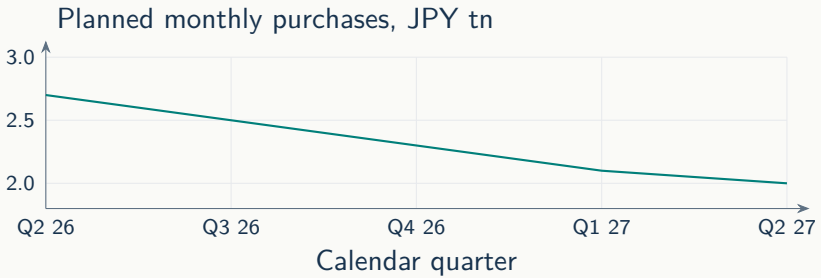
Japan policy evidence: September 2026

The 18 September BoJ decision set the overnight call-rate guideline at around 1.25%.

The new guideline took effect on 24 September.

A higher short rate is one component of normalisation.

Japan: the JGB purchase plan



BoJ plan announced 16 June 2026. Planned amounts, not realised purchases.

Why Japanese capital need not immediately return home

Domestic yields, hedging costs and expected FX returns all matter.

Hedged foreign bonds retain interest-rate and spread exposure.

Mandates, liabilities and existing hedges affect portfolio changes.

A hedge-cost illustration

$$R^{\mathrm{JPY}} = 1\%, \quad R^{\mathrm{USD}} = 4\%, \quad E = 150 \text{ JPY per USD.}$$

$$F = 150\frac{1.01}{1.04} = 145.67 \text{ JPY per USD.}$$

A covered one-year USD deposit returns 1% in yen under parity.

US/Japan counterfactuals

Shock relative to expectations	Conditional channel
BoJ path reprices higher	JPY carry funding becomes less attractive
JGB term premium rises	Long JGB prices fall at fixed cash flows
Global risk appetite collapses	Positions may unwind and premia change

Scenarios for reasoning, not predictions or descriptions of a realised event.

LECTURE 7

A macro trading lab

How does a conditional view become a sized position?

Markets react to the surprise in the full announcement

A 0.25-point cut was expected. A 0.50-point cut arrives.

The decision is 0.25 points more dovish than expected.

Guidance and economic news can change the later path too.

Exercise: a cut today, but a higher long yield?

After the announcement, assume:

Expected average short rate over ten years rises 0.10 points.

The ten-year term premium rises 0.20 points.

What happens to the ten-year yield? What about a duration-8 bond?

The decomposition gives the answer

$$\Delta R^{\text{spot}}(10) \approx 0.10 + 0.20 = 0.30 \text{ percentage points.}$$

$$\frac{\Delta P}{P} \approx -8 \times 0.003 = -2.4\%.$$

The long yield rises and the bond loses value in this specified scenario.

Turn a headline into a conditional chain

Identify the shock and the prior expectation.

State the model and policy response.

Translate the view into dated cash flows and yield changes.

Size the exposure and test the counterfactual.

Final exercise: a trade needs a falsifiable claim

Choose a shock, prior pricing and a policy response.

Predict two yields and one exchange rate, with assumptions.

Size the trade and name the counterfactual that would make it lose.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.